

Final Project Documentation — Automated Network Request Management

Date: 09 September 2026

Student Name: AVANAPU ROOPA

1. Introduction

Project Title: Automated Network Request Management in ServiceNow. A ServiceNow-based solution for capturing, routing, approving, and fulfilling network-related service requests.

2. Project Overview

Purpose: streamline and automate the end-to-end lifecycle of network-related service requests, reducing manual effort and enforcing standardized workflows.

Features: catalog-driven request intake, custom network request table, conditional UI Policies, an approval relationship linked to sysapprovals, and Flow Designer automation.

3. Architecture

The Service Portal and Network Request catalog item provide the requester experience. Catalog UI Policies and variables control interaction. Records are persisted in a custom network request table with related network configuration items, and approvals are tracked through a relationship to the sysapprovals table.

4. Setup Instructions

- Define business objectives and functional scope.
- Design the data architecture and create the custom network request table.
- Create the Approval Request relationship (System Definition > Relationships) against the sysapprovals table.
- Build the Network Request catalog item and configure ordered variables (100, 200, 300, ...).
- Configure Catalog UI Policies for device type and connection type scenarios.
- Build the "Network Request" flow in Flow Designer: Get Catalog Variables, Create Record, Ask for Approval, Send Email, Update Record.
- Configure access control and test the end-to-end request journey.

5. Configuration Structure

Catalog Item: Network Request. Data Table: custom network request table with standard platform tables.

Relationship: Approval Request (queries the sysapprovals table). Flow: Network Request (Flow Designer). Portal: Service Portal, accessed via the /sp URL prefix on the instance.

6. Running the Application

Log in to the ServiceNow instance, append the /sp prefix to the instance URL to reach the Service Portal, search for "Network Requests", complete and submit the catalog form, and confirm that a new request record and confirmation email are generated.

7. API Documentation

No custom server-side REST API was implemented. Catalog navigation uses standard Service Portal URL routing, and record operations are handled through the Flow Designer automation.

8. Authentication

Handled by the ServiceNow instance and Service Portal login configuration; access to the request table and approvals is governed by role-based access control.

9. User Interface

Network Requests catalog item on the Service Portal, dynamically-configured form variables, and conditional fields driven by UI Policies (e.g. Existing ID field, "Please specify" field).

10. Testing

Portal navigation, catalog variables, UI Policy conditional behavior, the Approval Request relationship, Flow Designer automation, and record creation were functionally tested. See the User Acceptance Testing document for detailed scenarios and results.

11. Screenshots / Demo

A walkthrough recording of the implementation is available at the following link:

https://drive.google.com/file/d/1gDrM3CmFP848MBk_zRy7qVAJv1ECrgzL/view?usp=sharing

12. Known Issues

No formal load benchmark has been performed. Advanced SLA-based escalation and CMDB-linked configuration item validation are outside the currently implemented scope.

13. Future Enhancements

- SLA-based escalation and automated reassignment.
- Deeper CMDB integration for network configuration items.
- Reporting dashboards for request volume, approval turnaround, and fulfilment time.
- Expanded approval hierarchies and delegation.
- Additional request types and knowledge-article recommendations during submission.